BiSem-Edge

Compute-Efficient Semantic Video with Complementary Reconstruction

Runcheng Zhang

Intelligent Networks and Communications Laboratory

KEY IDEA Retain BiSem's 4-keyframe / 64D interface; investigate compute redundancy, missing visual information and true CPU low-bit execution.

FIGURE 1 **System anatomy and measured sites of intervention**

All branches start from BiSem; improvements are evaluated independently

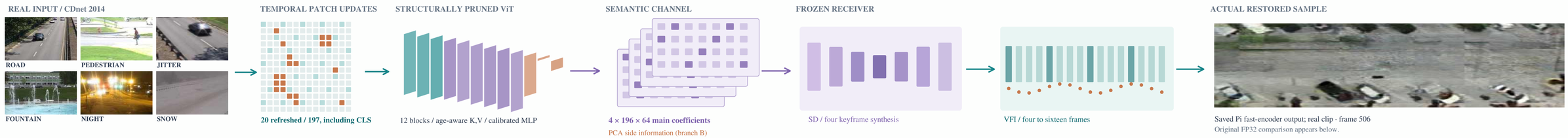


FIGURE 2 **Recorded reconstruction: preserving the reference is only part of the fidelity problem**



WHY ENCODER-ONLY ΔPSNR MISLEADS

Fast / FP32: −0.034 dB on this clip;
two outputs still look unlike the source.
A separate held-out CDnet pedestrian sample
has absolute reconstruction quality ~8 dB.
We therefore probe the 64D information loss.

01 / TEMPORAL EFFICIENCY

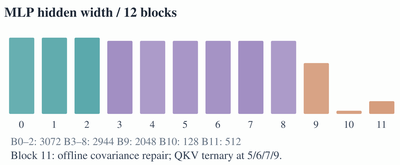
Spend computation on changed tokens

Reuse unchanged K/V states; keep global attention to cached positions.
Non-uniform MLP pruning and selective ternary QKV reduce CPU work.

A. Layered refresh, cached global context, physical pruning

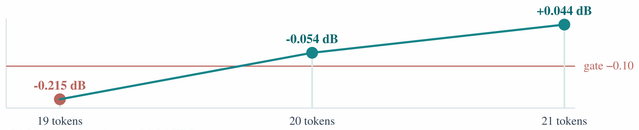


20 fresh positions
8 novelty + 12 novelty/age



B0–2: 3072 B3–8: 2944 B9: 2048 B10: 128 B11: 512
Block 11: offline covariance repair; QKV ternary at 5/6/7/9.

B. Minimum validated refresh budget (12 parking clips)



20 tokens: 56.90 FPS | 21 tokens: 56.36 FPS
The 19-token setting violates the frozen −0.10 dB threshold.

C. Cross-scene sender performance (batch-1; 12 cameras)

PTZ 73.9	SNOW 73.6	ROAD 74.7	OFFICE 75.2	WALK 75.2	JITTER 74.9
BOATS 75.3	OVERPASS 75.4	PARK 74.9	NIGHT 74.4	TRAM 75.0	SHADOW 74.6

74.78 source FPS macro mean (offline, 12,288 source frames).

D. Deployment trade-offs

20→32 updates: pedestrians 75.24→32.88 FPS; jitter 74.90→26.78.
2-clip batching: P95 wait 815→1,564 ms; scene jitter also hurts caching.

02 / REPRESENTATIONAL FIDELITY

What did 64 dimensions discard?

Diagnose appearance lost at the ViT 768→64 bottleneck.
Transmit nullspace-PCA coefficients; freeze BiSem's SD and VFI.

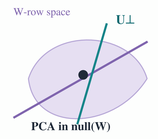
A. Diagnostic evidence at the 768→64 projection

Variance captured by 64 directions



Patch-RGB linear-readout probe / held-out cameras





B. Per-clip ΔPSNR across 12 held-out cameras

PTZ	SNOW	ROAD	OFF.	WALK	JITT.	BOATS	OVER.	PARK	NIGHT	TRAM	SHAD.
+1.10	-1.26	+0.15	+1.60	+3.73	+1.34	+0.51	+0.99	+2.88	-0.60	+0.22	+3.93
+0.86	-1.76	+0.13	+1.55	+3.75	+1.89	+0.35	+0.61	+3.41	-0.57	-0.19	+3.67

Camera-macro +1.180 dB | 19 / 24 clips improve | worst −1.759 dB.

C. Why complement? Matched decoder variants



Same KFS4, identical frozen SD/VFI; no new decoder training.

D. 64-bit/patch side packets, four-camera subset (8 clips)

PCA8 / 8 × INT8: **4.92 KB | 19.245 dB**
PCA12 / 4 × INT8 + 8 × INT4: **4.11 KB | 19.236 dB**
Packets −17% with −0.009 dB PSNR; no over-the-air test.
Residual motion / flicker is not consistently improved.

03 / NATIVE LOW-BIT EXECUTION

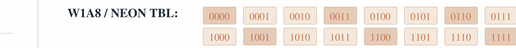
Make one-bit computation actually binary

Original BiLinear quantizes signs but still runs float GEMM.
Pack bits for Cortex-A76 NEON XNOR/CNT or 16-way LUT/TBL.

A. Integer dot products, not simulated quantization

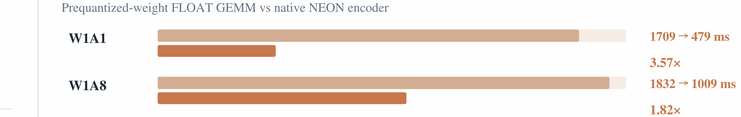


dot(a,w) = 2 · popcount(XNOR(a,w)) − K
Exact INT32 accumulation; recover token scale and layer bias.



B. Fixed-quantization baseline (4 keyframes / 4 threads)

Prequantized-weight FLOAT GEMM vs native NEON encoder



The rate includes activation packing; the GEMM comparator caches weights.

C. Operator shapes / Pi 5 microbenchmarks (ms)

ENCODER LINEAR	ACL FP32	W1A1	W1A8
SE QKV	20.98	**4.68**	15.03
Attention output	7.17	**1.82**	5.38
MLP up	27.90	**6.00**	19.83
MLP down	34.72	**5.36**	21.11

ACL FP32 differs in precision; table illustrates kernel shape costs.

D. Standalone packed state / numerical audit

Full state 458.01 → 31.37 MB **14.6×**
W1A8 full 80 layers: ΔPSNR ≤ 0.0054 dB; W1A1 up to 0.1648 dB.

EXPERIMENTAL DESIGN & READING GUIDE

Sender branch
Long-stream 64-clip parking comparison: 48.76 → 56.90 source FPS (Sep 11 → Sep 18).
Cross-camera 74.78 FPS is a separate batch-1, 25-FPS-replay evaluation.

Receiver branch
PCA directions/ridge fit on other cameras; CDnet 12×2 clips, 384 frames.
No actual mAP claim; small real-packet test excludes channel/FEC overhead.

CPU-native branch
Original frozen ViT/SD; 80 Linear layers; matched quantization/tiled kernel.
Timing applies to SE / SD only; VFI and camera-to-display delay excluded.